Supporting Information

Predicting aerobic granular sludge structural instability: An intelligent early-warning framework integrating convolutional neural network and fluorescence fingerprint features

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Content of Supplemental material

Number Supplementary pages: 36

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Number the Supplementary Tables: 10

Text S1. The characteristics of the reactor

Three identical laboratory-scale sequencing batch reactors (SBRs) constructed of plexiglass cylinders (height: 1100 mm; internal diameter: 70 mm; height-to-diameter ratio, H/D: 15.7; effective volume: 3 L) were utilized. Synthetic wastewater was fed into the reactors through submersible pumps at the top, with liquid level height controlled by a level relay. A drainage port positioned at the mid-height of each reactor was regulated by an electromagnetic drainage valve, achieving a 50% water exchange ratio (1.5 L per cycle). Fine-bubble aeration was provided via sand-cored micro-diffusers installed at the reactor base, with an air flow rate of 1 L/min maintained by glass rotameters. All reactors were operated at room temperature under a 6-hour cyclic regime comprising five phases: feeding, aeration, settling, drainage, and idling. Both feeding and drainage durations were fixed at 5 min, followed by a 30-min anaerobic phase after feeding. During the initial 7 days of operation, the settling time was progressively reduced from 15 min to 5 min, while the idling phase duration was maintained at 10 min. Operational timing for all phases was independently controlled by programmable timers (Liu et al., 2023).

Text S2. Inlet water condition for water quality shock test

The experimental influent water was artificially simulated wastewater system, the carbon source was prepared by sodium acetate, glucose and sodium propionate at the COD equivalent ratio of 1.8:1.2:2.5, the nitrogen source was chosen as ammonium sulfate, and the phosphorus source was used in the compounding of dipotassium hydrogen phosphate and potassium dihydrogen phosphate. The basic parameters of influent water were set as COD 550 mg/L, TN 45 mg/L, TP 7 mg/L, and the pH of the system was stabilized at about 8.0 by adding sodium bicarbonate buffer, meanwhile, trace metal elements such as $Cu^{2+}$and Mn^{2+}were supplemented. The specific ratios of each component are shown in Table. S3. (Vydehi et al., 2025).

(1)COD load shock experiment

In this stage, the dosage ratio of sodium propionate, glucose and sodium acetate was dynamically adjusted to achieve accurate regulation of COD load. The carbon source distribution parameters in the influent are shown in Table S4.

(2)Salinity shock experiment

The salt concentration of the influent was adjusted by adding NaCl according to the gradient. The amount of sodium chloride added according to different salt concentration gradients is shown in Table S4.

(3)pH shock experiment

HCl and NaOH solutions were used for dynamic adjustment of pH to construct acid-base shock conditions. The specific acid-base reagent addition scheme is shown in Table S4.

Text S3. High-throughput analytical methods for microbial communities

Total genomic DNA was extracted from sludge samples using the Ezup Column Soil DNA Extraction Kit from Sangon Biotech (Shanghai) Co., Ltd. The concentration and purity of the DNA were then checked using a Nanodrop2000 spectrophotometer, and the DNA samples that met the criteria were stored at -20°C. Sequencing was performed by Shanghai Personal Biotechnology Co.,ltd. (Shanghai, China) on the MiSeq platform (Illumina, USA). The upstream primer for bacterial sequencing was 338F (5'-ACTCCTACGGGAGGCAGCAG-3'), and the downstream primer was 806R (5'-GGACTACHVGGGTWTCTAAT-3'). For algal sequencing, the upstream primer was 3NDF (5'-GGCAAGTCTGGTGCCAG-3'), and the downstream primer was V4-euk-R2R (5'-ACGGTATCTRATCRTCTTCG-3') (Liu et al., 2022).

Text S4. Analysis of physicochemical indexes of sludge

(1) mixed liquor suspended solid(MLSS)

Quantitative filter papers were pre-dried in an oven at 103–105°C for 2 hours until constant weight was achieved, with the resultant weight recorded as m_0 (mg). A 100 mL mixed liquor sample was filtered through a Buchner funnel, during which the vessel walls were rinsed with deionized water to ensure complete transfer of sludge solids onto the filter paper. The filter paper containing the solids was subsequently dried in the oven for 2 hours. After drying, the filter paper was cooled in a desiccator and then weighed on an analytical balance, with the mass recorded as m1(mg). The mixed liquor suspended solids (MLSS, mg/L) were calculated according to eq. 2-1(Liu et al., 2024).

$$MLSS=\frac{(m_1-m_0)}{0.1} \tag{1}$$

(2) Sludge Volume Index after 30 minutes (SVI_{30})

SV30: During the aeration stage of the reactor, 100 mL of sludge-water mixture was taken from the reactor and placed in a 100 mL cylinder. After settling for 5 min, the volume of sludge was read as V_1 (mL), and after settling for 30 min, the volume of sludge was read as V_2 (mL). SV_{30} of sludge can be calculated according to eq. 2 respectively (Liu et al., 2024).

$$SV_{30}=\frac{V_2}{100}\times 100\% \tag{2}$$

According to eq. 3, SVI30 of sludge can be calculated separately

$$SVI_{30}=SV_{30}/MLSS\times 10^6 \tag{3}$$

In addition, the average granule size of sludge in this study was measured by laser granule size distribution tester, and the morphology change of sludge was observed by Nikon optical microscope.(Liu et al., 2024).

Text S5. ResNet Introduction and Parameter Configuration

ResNet (Residual Neural Network), a pivotal breakthrough in the field of convolutional neural networks (CNNs), was proposed by He Kaiming's team in 2015. By introducing a residual learning framework, the vanishing gradient and model degradation problems in deep networks were effectively resolved. Its core structure incorporates skip connections within residual blocks, enabling direct propagation of input features to deeper output layers ($H(x) = F(x) + x$), thereby significantly enhancing network trainability. Depending on network depth, two residual block designs were adopted: the basic residual block composed of two 3×3 convolutional layers (for shallow networks such as ResNet-18/34), and the bottleneck residual block implementing dimensionality reduction and expansion via a $1\times1 \rightarrow 3\times3 \rightarrow 1\times1$ convolutional sequence (for deep networks including ResNet-50/101). This design maintained feature representation capability while reducing parameter complexity.(Altarez, 2024)

In this study, four depth variants—ResNet-18, 34, 50, and 101—were comparatively analyzed. All architectures were optimized using SGD (initial learning rate: 0.1, momentum: 0.9), with transfer learning, batch normalization (batch size：18), and data augmentation employed to balance training stability and generalization capability. This study implements a comprehensive global fine-tuning scheme in ResNet transfer learning without pre-frozen layers. By enabling end-to-end joint optimization across all feature levels from low-level to high-level on the target domain, the model overcomes the limitations of initial pre-trained representations. It actively reconstructs its representation space to achieve optimal alignment with the task-specific data distribution, thereby avoiding representation bias and model capacity waste caused by hierarchical freezing. In the process of model training, in order to eliminate the influence of randomness on the training results, the random seed of model training process is set to 52. The model runs on PyTorch version 2.8.0 with cu128, CUDA 12.8, and cuDNN 91002. It was trained on an NVIDIA GeForce RTX 5080 Laptop GPU for 5 hours.

Text S6. Model performance metrics

Since the EPS-ResNet model in this study is a classification model. The performance of each developed model was evaluated using three model performance metrics analysed as Accuracy, F1, Precision and Recall. The formulas for these metrics are as follows (Zhu et al., 2023). Eq. (6-8) (Jeon et al., 2024).

$$\text{Accuracy}=\frac{\text{TP+TN}}{\text{TP+TN+FP+FN}} \tag{4}$$

$$\text{Precision}=\frac{\text{TP}}{\text{TP+FP}} \tag{5}$$

$$\text{Recall}=\frac{\text{TP}}{\text{TP+FN}} \tag{6}$$

$$\text{F1}=2\times\frac{\text{Accuracy}\times\text{Precision}}{\text{Accuracy+Precision}} \tag{7}$$

Where TP: is predicted as positive is correct, FP: is predicted as positive is wrong, FN: is negative is wrong, TN: is predicted as negative is correct.

The visualization of model performance was addressed through the adoption of a confusion matrix. As a fundamental and intuitive tool in machine learning for evaluating classification model performance, the Confusion Matrix employs an N×N table structure (where N denotes the number of classes) to illustrate the correspondence between predictions and actual conditions: the rows of the matrix represent the true classes of the data, while the columns represent the predicted classes from the model. Its key value lies in clearly revealing correct classifications by the model (e.g., elements on the diagonal indicate the number of correctly predicted samples) and various types of errors (off-diagonal elements), such as misclassifying actually positive samples as negative (False Negative, FN) or negative samples as positive (False Positive, FP). Based on these fundamental data, key evaluation metrics including Accuracy, Precision, Recall, and F1-score were calculated, enabling a comprehensive assessment of the model's strengths and weaknesses, identification of confusions or biases towards specific classes, and providing clear direction for subsequent model optimization (Zhu et al., 2023).

Text S7. 5 Biomarker Characterization Regions for Excitation-Emission-Matrix Spectra

Based on fluorescence characteristics at distinct excitation-emission wavelengths, Excitation-Emission-Matrix Spectra (EEMs) spectra were categorized into five characteristic regions: Region I (Ex: 200–250 nm, Em: 200–330 nm), characterized by tyrosine-like simple aromatic proteins; Region II (Ex: 200–250 nm, Em: 330–380 nm), corresponding to tryptophan-like proteins; Region III (Ex: 200–250 nm, Em: 380–550 nm), representing fulvic acids, quinones, and phenolic compounds; Region IV (Ex: 250–550 nm, Em: 200–380 nm), indicative of soluble microbial metabolites; and Region V (Ex: 250–550 nm, Em: 380–550 nm), associated with high-molecular-weight aromatic organics including humic acids, humic-like substances, and polycyclic aromatic hydrocarbons (Zhang et al., 2025).

Text S8. Association analysis method based on Mantel test

Mantel test, a non-parametric statistical method based on permutation tests, was proposed by Nathan Mantel in 1967 to evaluate correlations between two distance matrices. This method quantified the association by calculating standardized correlation coefficients (Pearson coefficient) between matrices (e.g., species distribution vs. environmental factor matrices), with statistical significance (p-values) derived from repeated random permutations of sample orders. Its core advantage lay in handling spatially autocorrelated data while circumventing the independent sample assumption required by traditional correlation coefficients, rendering it applicable to multivariate association analyses in ecology, microbial ecology, and environmental sciences. In this study, Mantel test was implemented via the R programming language to assess correlations among five fluorescence regions, biological community structures, and functional gene profiles (Liu et al., 2024).

For the quantification of EPS-related EEMs image data, this study employed a weighted average method to calculate the fluorescence intensity of each pixel across the five fluorescent regions of the measured EEMs,thereby quantifying the intensity of each EEM's five fluorescent regions. For microbial data, the relative abundance determined by 16S sequencing was directly used to characterize each biomass.

Text S9.Occlusion Sensitivity Analysis

Occlusion Sensitivity Analysis was defined as a visualization technique employed to interpret the decision-making mechanisms of neural networks. Its core concept lay in systematically occluding different regions of an input image and observing the corresponding changes in the model's prediction probability or output error, thereby identifying which areas were most critical for the model's decision. This method typically involved sliding an occlusion patch (a white square) across the image and sequentially calculating the decline in prediction probability or the increase in output error when specific regions were obscured; the final results were commonly presented as a heatmap, wherein high-sensitivity areas (those where occlusion led to a significant probability drop or error increase) indicated regions containing features vital for the model's classification or regression tasks. In the field of computer vision, this approach was not only utilized to evaluate whether image classification models relied on correct target regions but also to analyze the robustness of models to partial occlusion in complex tasks, revealing how models leveraged information from visible parts to infer occluded sections.

Text S10. Method of Extracting Feature Maps of Residual Modules

The extraction of features from key residual layers in the EPS-ResNet model in this study was implemented in a Python environment, leveraging the PyTorch deep learning framework along with the Matplotlib library to visualize feature maps of the residual modules. The procedure was conducted as follows: a pre-trained model was first loaded and set to evaluation mode; subsequently, the output tensors of specified residual layers were captured using a forward hook approach. The forward hook, facilitated by the register_forward_hookfunction, dynamically intercepted intermediate layer features without altering the model architecture. Specifically, the input images underwent preprocessing (including resizing and normalization), model forward propagation was executed to acquire the feature maps, and finally, the feature tensor was converted into an image format and visualized using Matplotlib's imshowfunction, enabling an intuitive analysis of the feature extraction process from shallow textures to deep semantic representations.

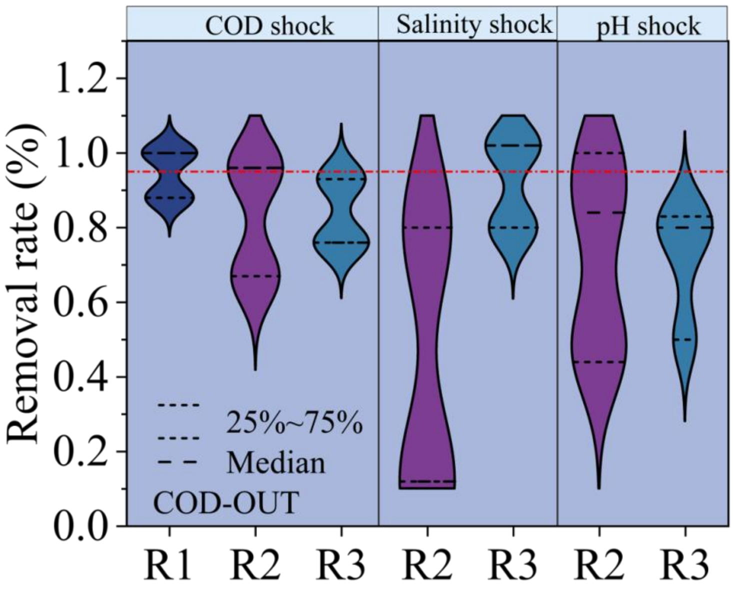


Fig. S1. Characterization of decontamination of AGS system under different water quality shocks. COD removal under three water quality shocks.

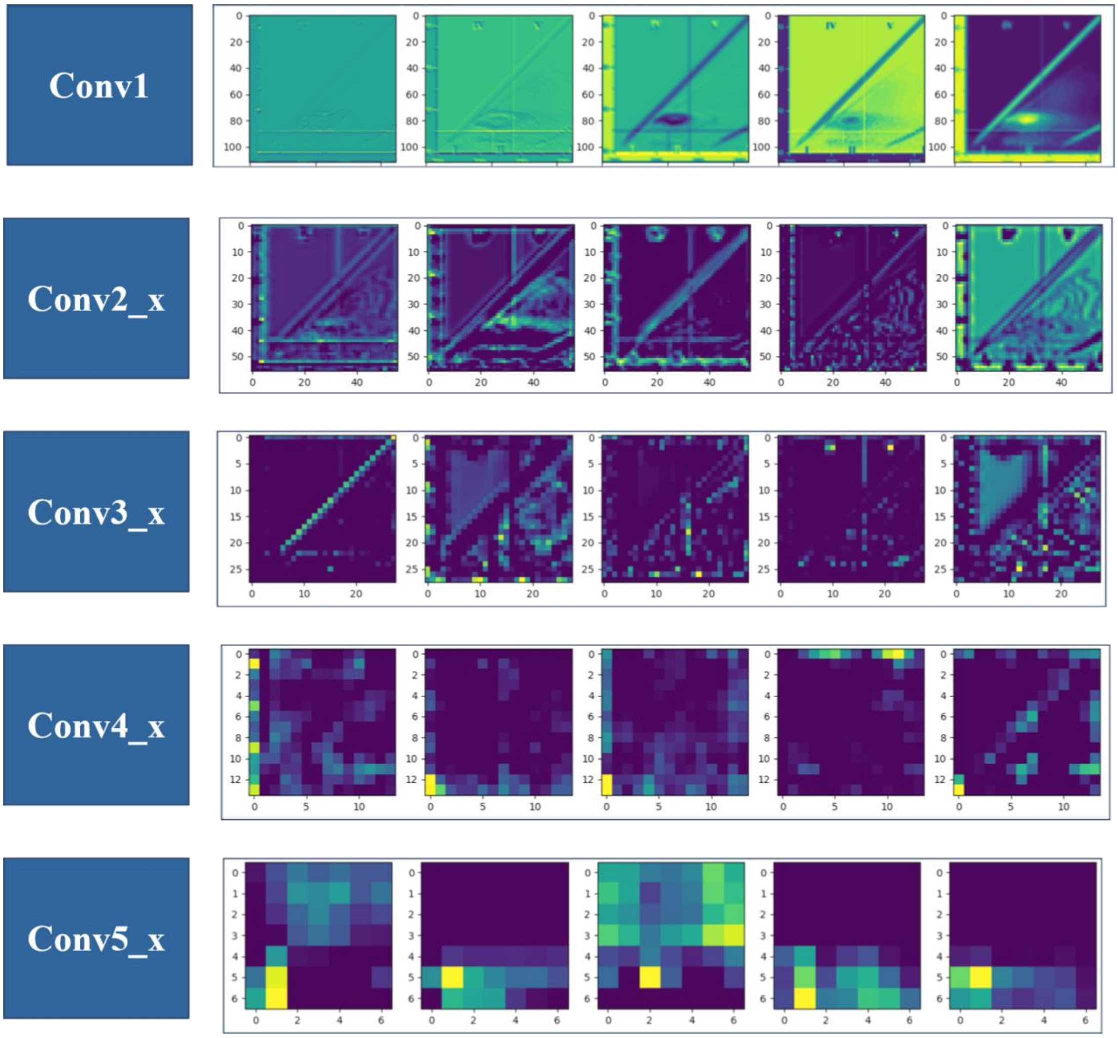


Fig. S2. Map of features extracted from each residual module of the LB-ResNet model (partial)

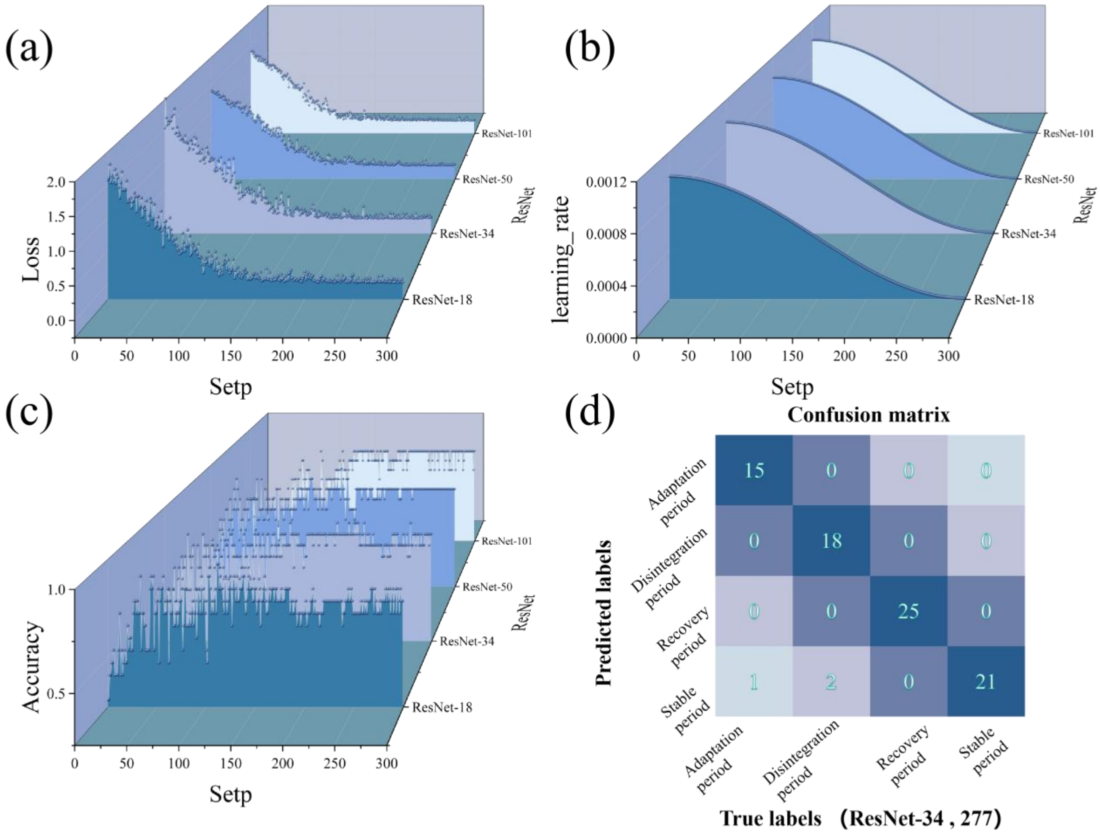


Fig. S3. AGS granular structure instability warning model (TB-ResNet) based on the input of EEMs from TB-EPS, (a) the loss rate during the training of the four ResNet structure models; (b) the learning rate during the training of the models; (c) the accuracy rate of the test set during the training of the models; and (d) the optimal AGS instability warning selected by combining the ratio of the accuracy rate of the models to the generalization ability of the model.

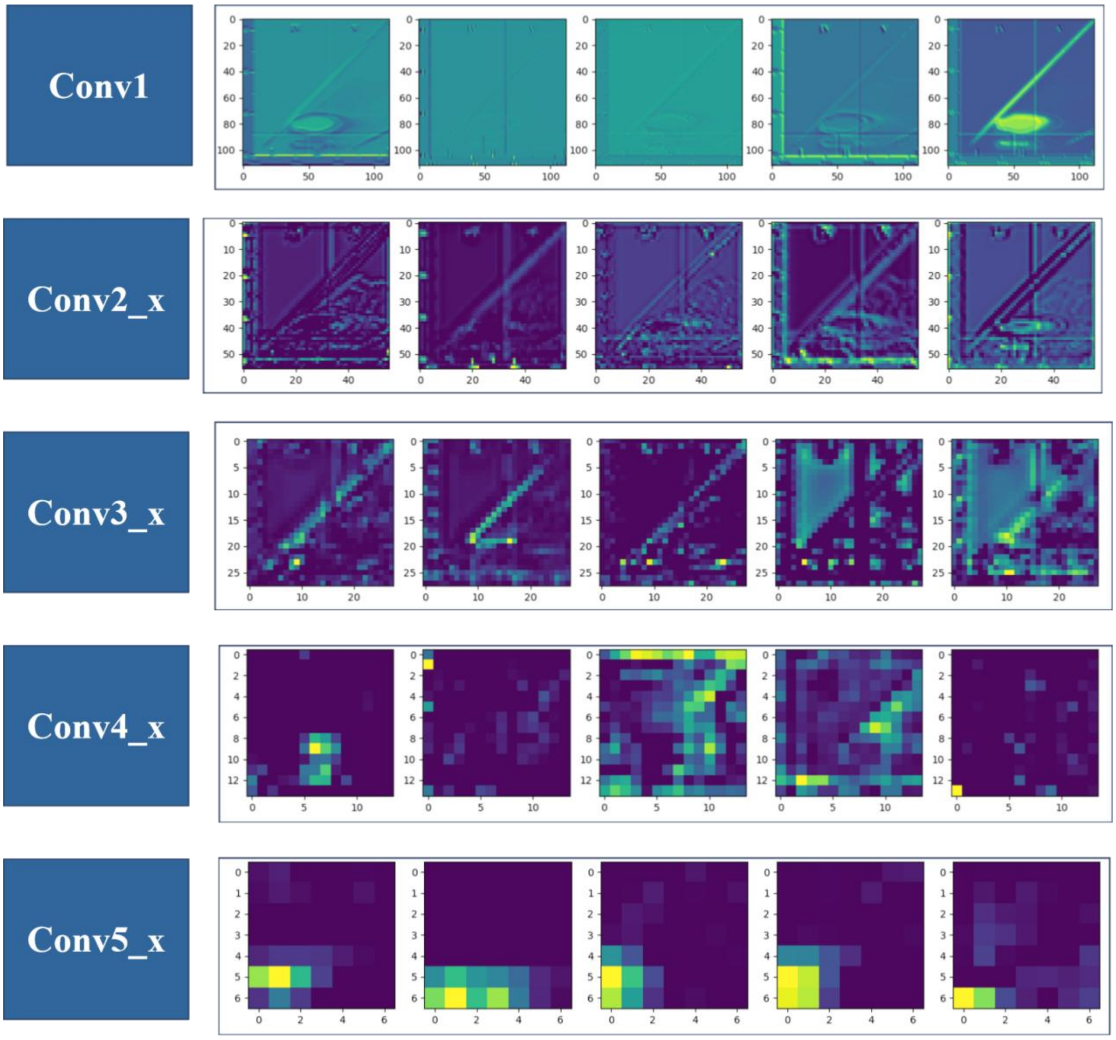


Fig. S4. Map of features extracted from each residual module of the TB-ResNet model (partial)

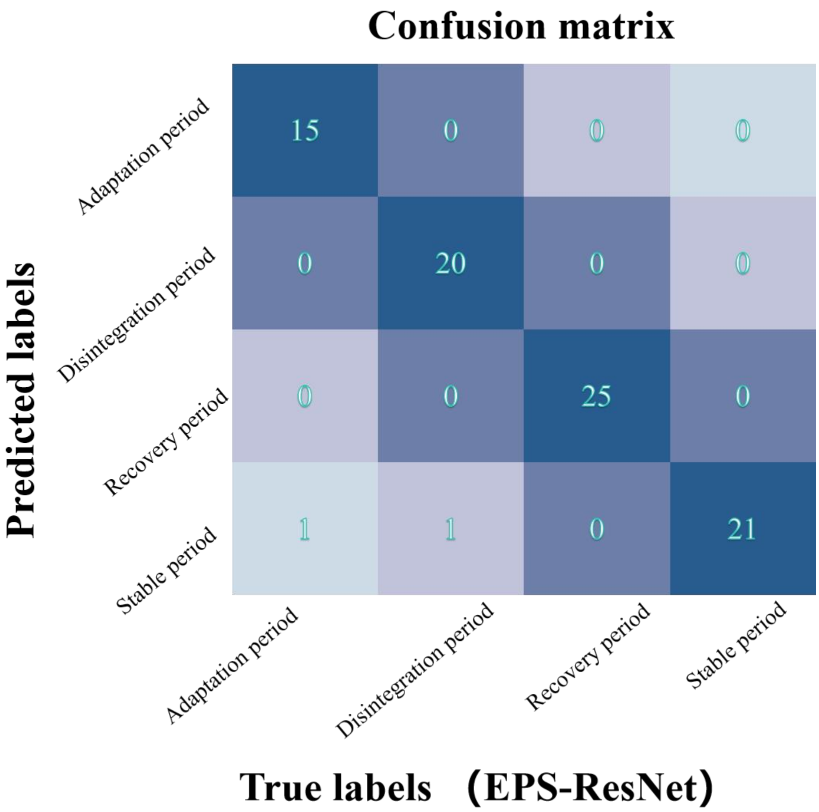


Fig. S5. Confusion matrix results for the EPS-ResNet model

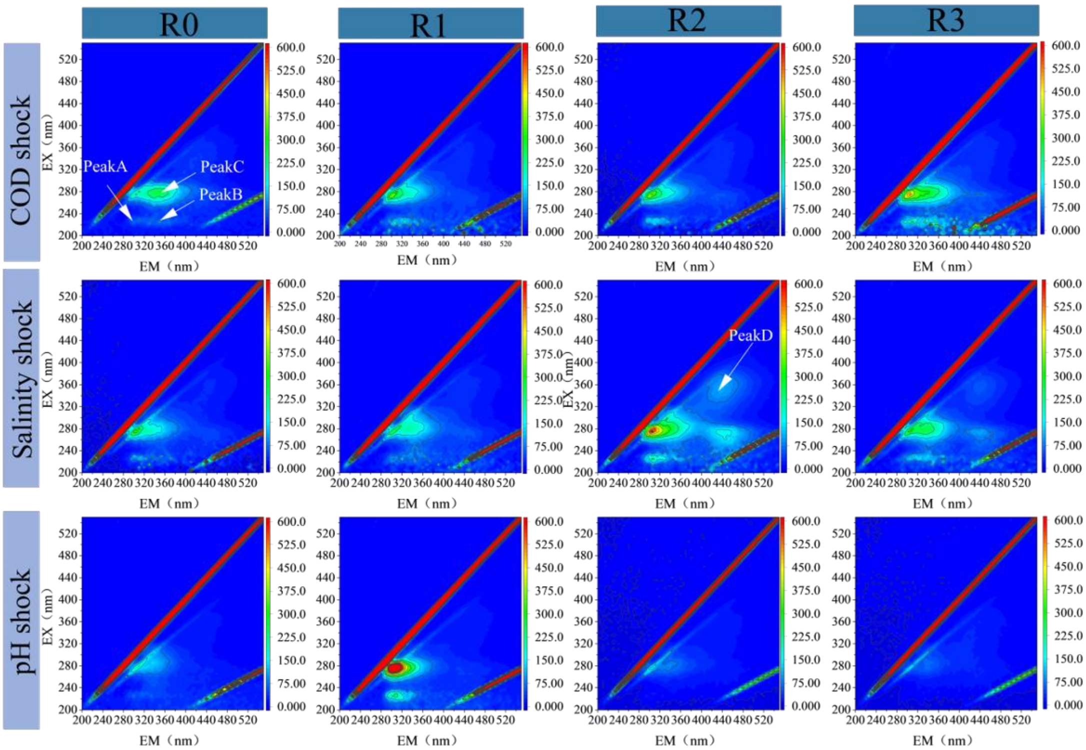
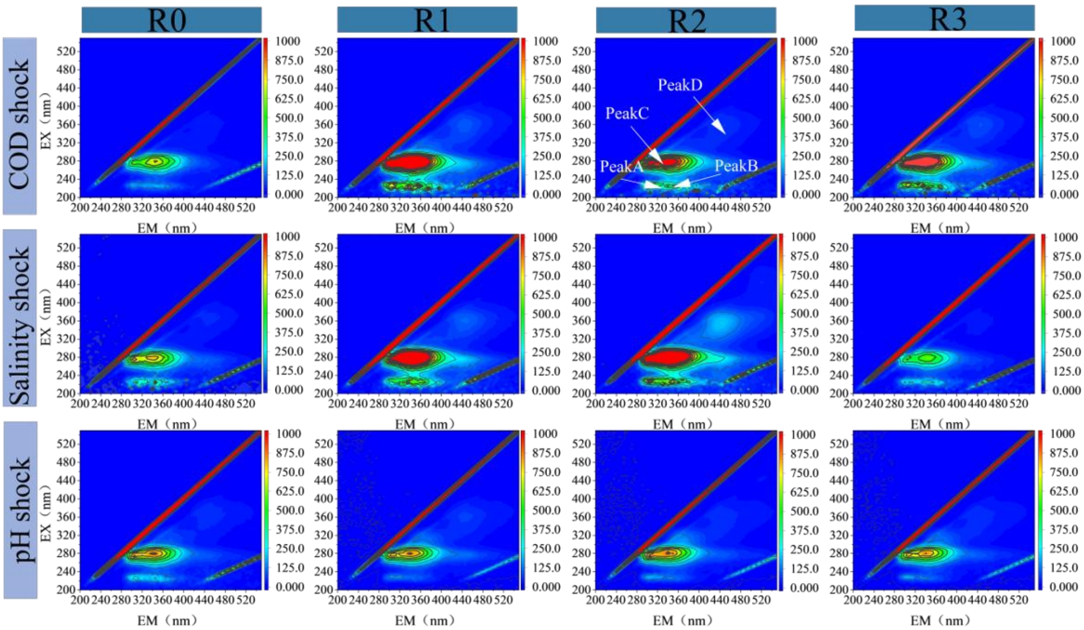


Fig. S6. Characterization analysis of LB-EPS EEMs



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250 **Fig. S7.** Characterization analysis of TB-EPS EEMs

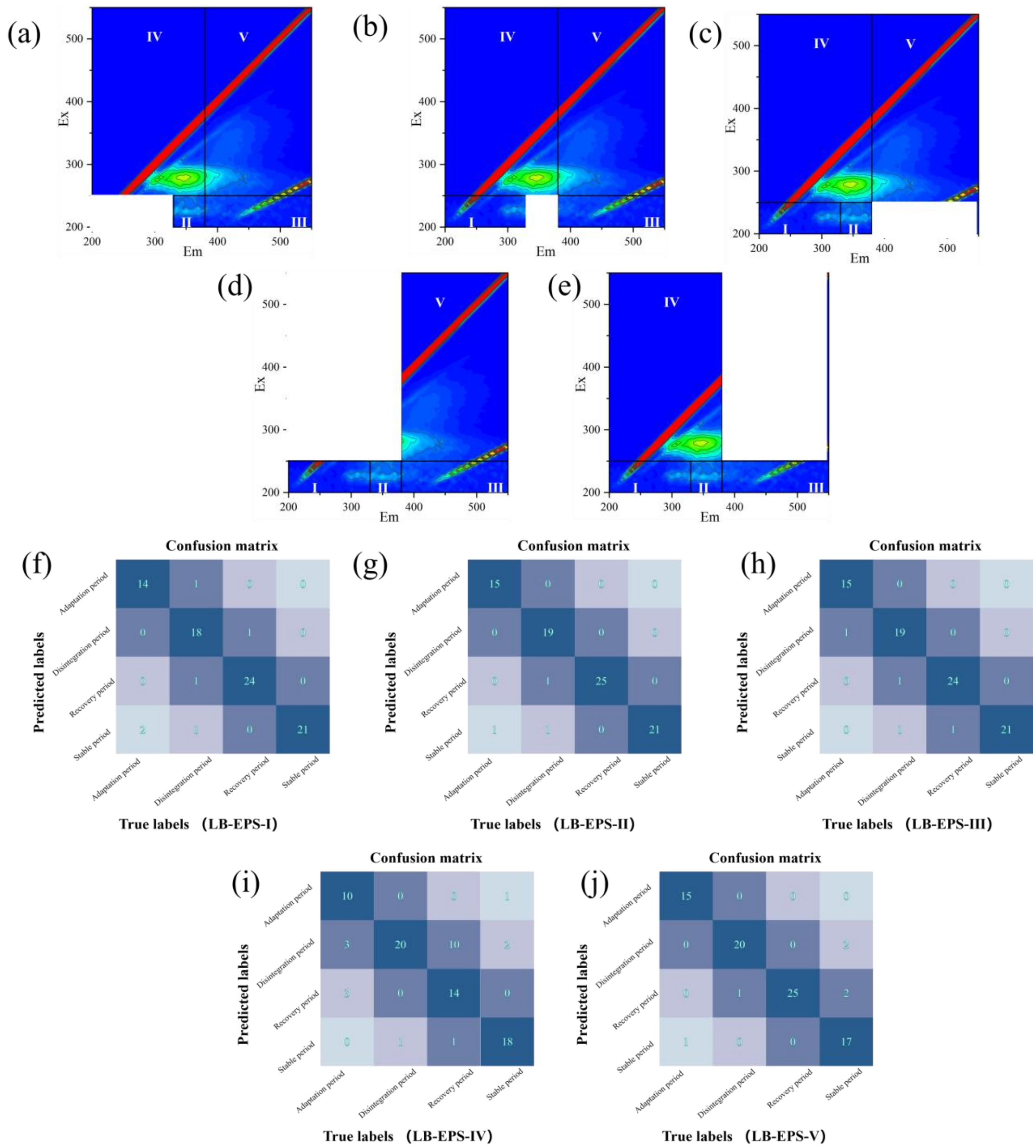

Fig. S8. Contribution analysis of the five regions of the EEMs of LB-EPS, (a)-(e) Masking effect of the fluorescent regions from I to V, (f)-(j) Performance of the LB-ResNet model after masking of the fluorescent regions from I to V

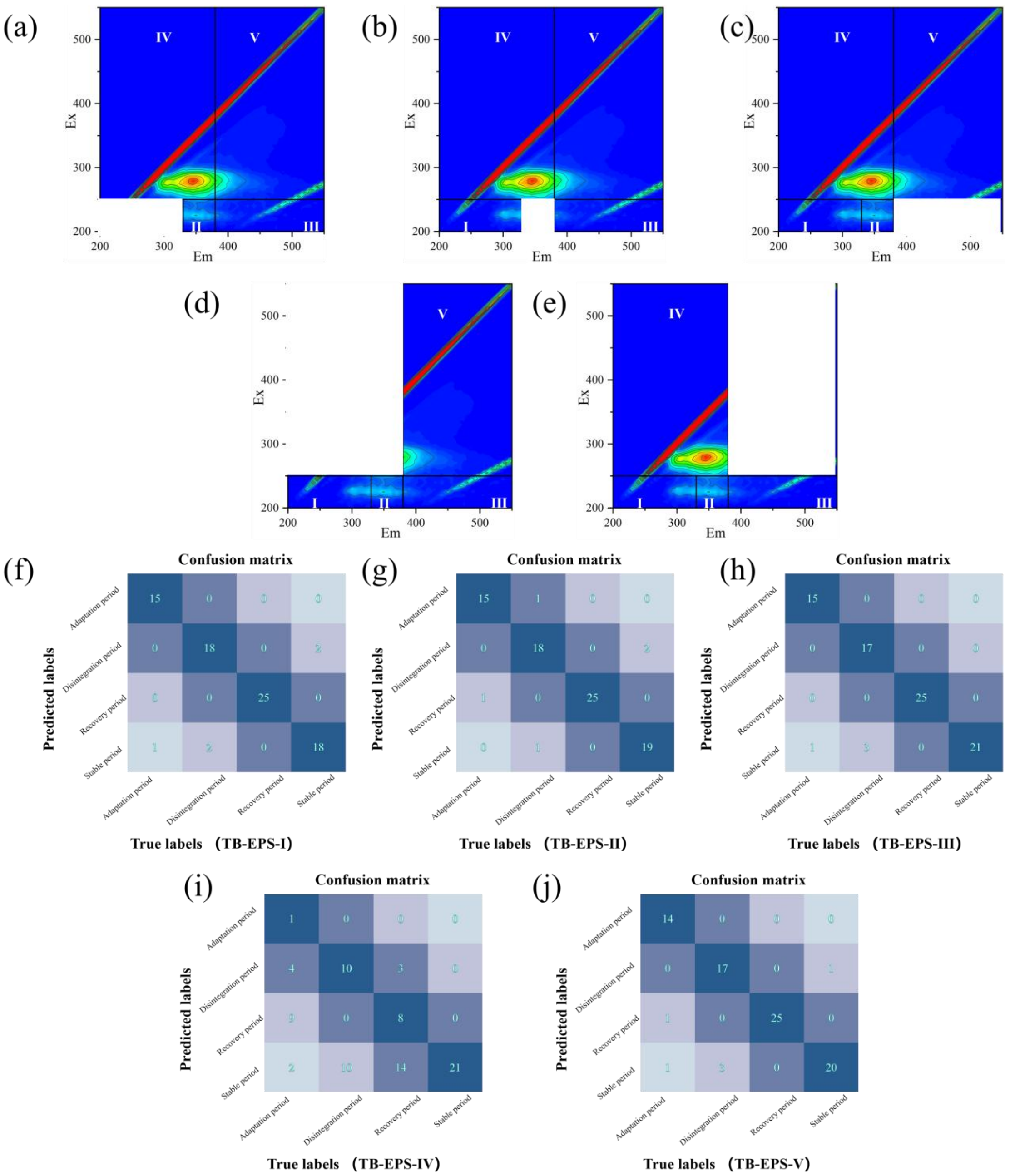

Fig. S9. Contributi on analysis of the five regions of the EEMs of TB-EPS, (a)-(e) Masking effect of the fluorescent regions from I to V, (f)-(j) Performance of the TB-ResNet model after masking of the fluorescent regions from I to V.

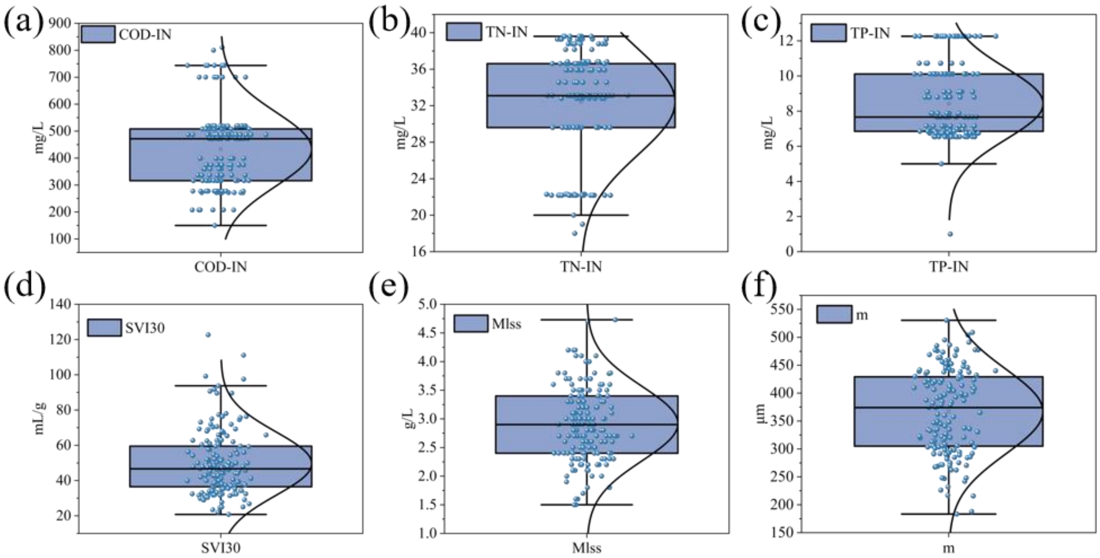

Fig. S10. Distribution analysis of measured sample data and anomaly detection based on the quartile method. (a) COD-IN, (b) TN-IN, (c) TP-IN, (d) SVI30, (e) MLSS, (f) m. Note: Values outside the upper and lower bounds are outliers and have been excluded.

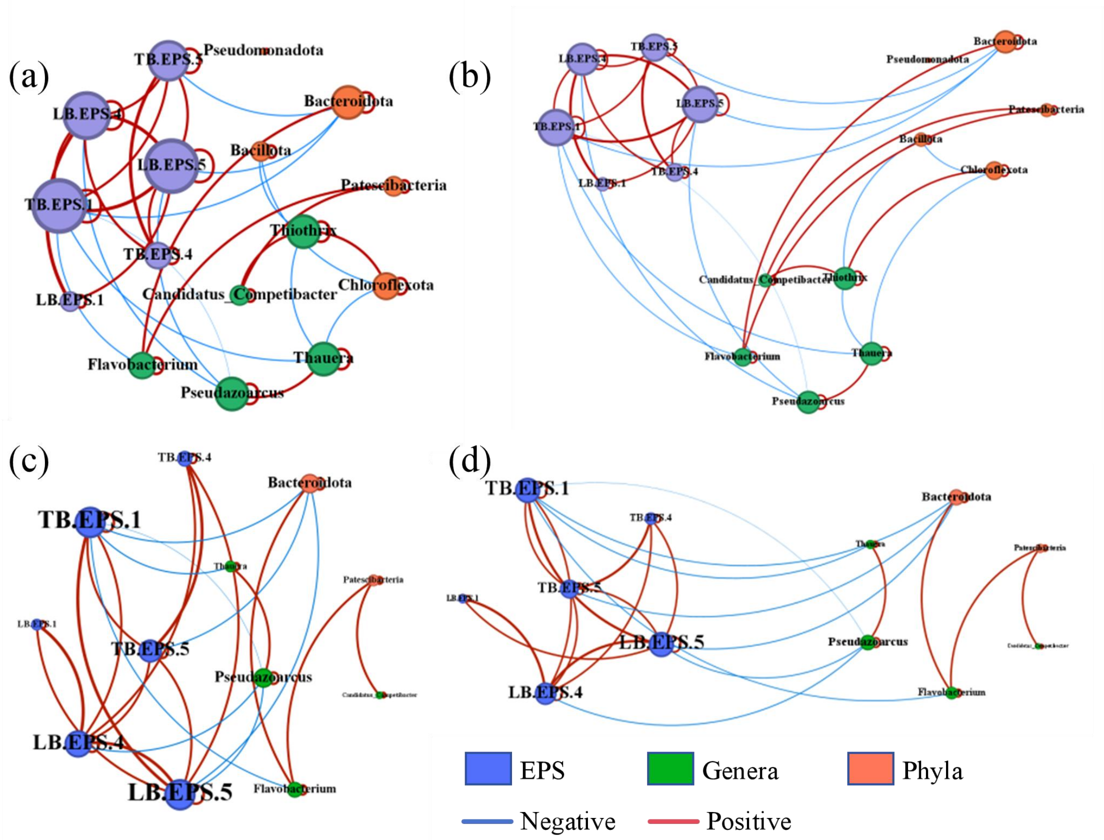

Fig. S11. Co-occurrence network diagram of key fluorescent regions and critical microbial genera in EPS. (a)-(b) Full network diagram, where node size indicates degree of connectivity. (c)-(d) Key subnetworks, highlighting core interactions most relevant to model early warning performance.

Table S1

Descriptions of the features present in the dataset used for modeling.

Parameter	Description
Time (t) (day)	SBR reactor incubation time for AGS
COD-IN(mg/L)	Chemical Oxygen Demand content in SBR reactor feed water
TN-IN(mg/L)	Total nitrogen in SBR reactor feed water
TP-IN(mg/L)	Total phosphorus in SBR reactor feed water
MLSS(g/L)	Mixture liquid suspended solid in the feed water to the SBR reactor
SVI_{30}(mL/g)	Sludge volume index of sludge in SBR reactors
m(μm)	Granule size of AGS in SBR reactor
COD-OUT(%)	COD removal for one cycle in a SBR reactor
LB-ResNet	AGS instability warning model based on EEMs inputs from LB-EPS
TB-ResNet	AGS instability warning model based on EEMs inputs from TB-EPS
EPS-ResNet	AGS instability warning model based on two-view inputs of EEMs from LB-EPS and TB-EPS
LB-EPS-X (X=ItoV)	Correlation analysis of X-fluorescence regions of EEMs of LB-EPS
TB-EPS-X (X=ItoV)	Correlation analysis of X-fluorescence regions of EEMs of TB-EPS

Table S2

Scope and Data Source of Key Indicators in Model Training Set

Parameter*	MLSS(g/L)	SVI_{30}(mL/g)	m(μm)	COD-IN(mg/L)	TN-IN(mg/L)	TP-IN(mg/L)
Min	1.50	20.78	183.30	207.66	22.17	6.54
Max	4.73	122.67	530.50	743.66	45.42	12.26
Parameter*	COD-OUT(%)	Data volume and data sources				
Min	12.00	Training set	332	Data sources,	Laboratory self-collection	
Max	99.00	Test set	83	Experiment condition	See Text S2	

*The full name and interpretation of each acronym variable are given in Table S1.

Table S3

Formula for artificial simulated inflow

Basic elements	Content ($mg \cdot L^{-1}$)	Trace element	Content($mg \cdot L^{-1}$)
$(NH_4)_2SO_4$	180.0	H_3BO_3	150
$NaHCO_3$	180.0	$CoCl_2 \cdot 6H_2O$	150
KH_2PO_4	21.0	KI	180
K_2HPO_4	32.0	$MnCl_2 \cdot 4H_2O$	120
$MgSO_4.7H_2O$	12.0	$ZnSO_4 \cdot 7H_2O$	120
$CaCl_2$	10.0	$Na_2MOO_4 \cdot 2H_2O$	60
$FeSO_4 \cdot 7H_2O$	16.4	$CuSO_4 \cdot 5H_2O$	30
CH_3COONa	230.0	$FeSO_4 \cdot 7H_2O$	10
CH_3CH_2COONa	180.0	$CaCL_2 \cdot 2H_2O$	10
$C_6H_{12}O_6$	110.0	EDTA	10
Trace element	1.0 mL/L		

Table S4

Intake agent dosing for water quality shocks

	Dosing components	content ($mg \cdot L^{-1}$)		
		Normal water intake(R1)	Under COD load(R2)	High COD load(R3)
COD shock	Sodium propionate	180.0	360	90
	Glucose	110.0	220	55
	Sodium acetate	230.0	460	115
Salinity shock	Dosing components	Normal water intake (R1)	0.77(mmol/L) salinity	1.54(mmol/L) salinity
	Sodium chloride	0	45	90
pH shock	Dosing components	Normal water intake (R1)	pH =5±0.1	pH=9±0.1
	HCL/NaOH	0	0.3	3.33

Table S5

Evaluation index of LB-ResNet model

LB-ResNet*	Adaptation period	Disintegration period	Recovery period	Stable period
Precision	1.00	1.00	0.96	0.91
Recall	0.94	0.90	1.00	1.00
F1	0.97	0.95	0.98	0.95
Accuracy	0.964			

*The full name and interpretation of each acronym variable are given in Table S1.

Table S6

Evaluation index of TB-ResNet model

TB-ResNet*	Adaptation period	Disintegration period	Recovery period	Stable period
Precision	1	1	1	0.88
Recall	0.94	0.90	1	1
F1	0.97	0.95	1.00	0.94
Accuracy	0.963			

*The full name and interpretation of each acronym variable are given in Table S1.

Table S7

EPS-ResNet model evaluation metrics

EPS-ResNet*	Adaptation period	Disintegration period	Recovery period	Stable period
Precision	1.00	1.00	1.00	0.91
Recall	0.94	0.95	1.00	1.00
F1	0.97	0.97	1.00	0.95
Accuracy	0.976			

*The full name and interpretation of each acronym variable are given in Table S1.

Table S8

Quantitative indicators (precision and recall) for the contribution analysis of the five regions of the EEMs of the LB-EPS

Evaluation index	Fluorescence region*	Adaptation period	Disintegration period	Recovery period	Stable period
Precision	LB-EPS-I	0.07	0.05	0.00	0.04
	LB-EPS-II	0.00	0.05	0.00	-0.04
	LB-EPS-III	0.00	0.05	0.00	0.00
	LB-EPS-IV	0.09	0.43	0.14	0.01
	LB-EPS-V	0.00	0.09	0.07	-0.03
Recall	LB-EPS-I	0.06	0.05	0.04	0.00
	LB-EPS-II	0.00	0.00	0.00	0.00
	LB-EPS-III	0.00	0.00	0.04	0.00
	LB-EPS-IV	0.31	-0.05	0.44	0.14
	LB-EPS-V	0.00	-0.05	0.00	0.19

*The full name and interpretation of each acronym variable are given in Table S1.

Table S9

Quantitative indicators (precision and recall) for the contribution analysis of the five regions of the EEMs of the TB-EPS

Evaluation index	Fluorescence region*	Adaptation period	Disintegration period	Recovery period	Stable period
Precision	TB-EPS-I	0.06	0.10	-0.04	0.06
	TB-EPS-II	0.06	0.10	0.00	-0.04
	TB-EPS-III	0.00	0.00	-0.04	0.07
	TB-EPS-IV	0.00	0.41	0.49	0.47
	TB-EPS-V	0.00	0.06	0.00	0.08
Recall	TB-EPS-I	0.00	0.00	0.00	0.14
	TB-EPS-II	0.00	0.00	0.00	0.10
	TB-EPS-III	0.00	0.05	0.00	0.00
	TB-EPS-IV	0.88	0.40	0.68	0.00
	TB-EPS-V	0.06	0.05	0.00	0.05

*The full name and interpretation of each acronym variable are given in Table S1.

Table S10

Accuracy of EEMs of LB-EPS and TB-EPS after 5-region masking

Fluorescence region*	Accuracy	Fluorescence region	Accuracy
LB-EPS	0.964	TB-EPS	0.963
LB-EPS-I	0.928	TB-EPS-I	0.927
LB-EPS-II	0.964	TB-EPS-II	0.939
LB-EPS-III	0.952	TB-EPS-III	0.951
LB-EPS-IV	0.747	TB-EPS-IV	0.488
LB-EPS-V	0.928	TB-EPS-V	0.927

*The full name and interpretation of each acronym variable are given in Table S1.

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Environmental Machine-learning, Baseline Reporting, And Comprehensive Evaluation

The EMBRACE Checklist

Version 1.0. Last Updated Date: September, 2024.

The main purpose of EMBRACE Checklist is to facilitate the reporting of basic data and methods used for supervised machine learning (ML) research in environmental science and engineering (ESE). It also aims to provide a comprehensive evaluation of important preparation and analysis steps. We warmly encourage researchers to include the checklist as a reference, and if interested list it as supplementary materials in their publications.

Detailed instructions and additional resources are available open access for download from the author's GitHub: https://github.com/starfriend10/EMBRACE

- Future updates and new resources will be available in the above GitHub repository, which is also served as a community-owned platform to share data and codes, refine forms/checklists, discuss ML-ESE topics, promote research outcomes, and continuously identify/avoid pitfalls and follow good practices.
- We encourage you to direct share your checklists, but you can also save it as a read-only document. See instructions for detailed steps.
- Please cite the Viewpoint when using the checklist. If you have any questions or suggestions, please contact Dr. Jun-Jie Zhu at Princeton University (junjiez@princeton.edu or ranmuweijie@gmail.com).

Project Overview			
Project Title			
Contributing Authors			
Date		Completed by	
Contact Email		DOI (if published)	
Domain Category		Or Specify:	
Learning Type	Regression	Classification	Regression+ Classification
Prediction Type	Deterministic	Probabilistic	Deterministic+ Probabilistic
Other Information			

I. Problem Formulation							
Project Objective(s)							
Feasibility Assessment	Data Availability	Model Accessibility	Computation Resources	Knowledge Preparation	Time Availability	Financial Availability	Risk Tolerance
Levels							

II. Data Collection								
Data Sources		Time-series Monitoring		Field Campaigns		Government Database		Scientific Literature
		Laboratory Experiments		Simulation Outputs		Others. Specify:		
Source Types		Ordinary		Time-series		Literature-based		Others. Specify:
Data Types		Continuous Numerical		Categorical		Textual		Visual
		Graphical		Auditory		Others. Specify:		
Data Ethics		No special ethical considerations are required, or followed the necessary requirements when needed						
		No special permissions are required, or obtained the necessary permissions when needed.						
		Provided appropriate credit(s) to the data source(s)						
Data Availability		Data are publicly available through a PURL or DOI		Data are available as requested				
Raw Data Quantity	Raw Sample Size		Raw Variable Size		Raw Total Data volume		Raw Sample-Variable Ratio	
Raw Data Quality	The raw data had an internal QA/QC before data retrieval?				Yes		No	
	The raw data went through a peer-review or similar process?				Yes		No	

III. Data Preprocessing									
Data Cleaning		Abnormal Data Management		Statistical Outlier Management		Technical Outlier Management			
		Data Correction due to Technical Limitations		Others. Specify:					
Data Enrichment		Missing Data Management		Data Augmentation		Data Aggregation		Data Balancing	
	Upper sample ratio among groups:		Others. Specify:						
Feature Engineering		Initial Variable(s) Exclusion		Feature Importance Analysis		Feature Extraction			
		Categorical Feature Transformation		Dummy Feature Removal					
		Ordinal Encoding	Encoded variable(s):						
		Feature Selection		Feature Scaling (for X)		Feature Scaling (for Y)		Circular Feature Scaling	
		New Feature Development		Ordinary Mathematical Transformation(s)		Time-Dependent Feature Transformation(s)			
				Spatial Feature Transformation(s)		Others. Specify:			
	Sparse Dataset?	Yes	No	Preprocessed to increase information density?	Yes	No			
		Normality Test (before/after transformation)		Others. Specify:					
Data Splitting	Framework		Training-Testing		(Pre-)Training-Validation-Testing		Cross-Validation-Testing		
	Splitting Method		Simple Random		Grouped Random		Simple Block		Sub-Group R/B
		Ensured training dataset covered most possible scenarios (e.g., different seasons)							
Final Data Quantity	Testing Ratio	Preprocessed Sample Size	Preprocessed Feature Size	Sample-Feature Ratio	Training SFR				

IV. Methods Selection and Model Initialization							
Methods Selection		Selected supervised learning methods to be tested based on case needs (e.g., data, computation resources, etc.)?					
	Methods Studied: N ML and M Statistical	N =	M =				
	ML Family		Tree-based		Neural Network		Others
	ML Type		Shallow ML		Deep ML	Final Optimal Method(s):	
Model Uncertainty		Examined Random Seeds		Replicated Modeling Process			
		Assessed probabilistic prediction results		Others. Specify:			

V. Model Evaluation and Model Optimization										
Model Evaluation		Set a goal model performance before evaluation								
	Specify Evaluation Metrics:									
		Studied under-fitting issue		Studied over-fitting issue						
		Final evaluation was based on the testing data		Weighted metrics were used for imbalanced data						
		ML significantly outperformed statistical models		Statistical perform similar or even better than ML models						
Hyper-parameter Optimization	HPO Method		No HPO		Trial-and-Error/Experience		Grid/Random Search (similar)		Metaheuristics	
			Other HPO. Specify:							
	Number of methods tuned by HPO:	Number of hyperparameters searched:								
	Number of values in each hyperparameter:		Continuous Search		Discrete Search. Specify the number:					
	CV Method		No CV		k-fold CV		Leave-one-out CV		Stratified CV	
			Pre-defined or other CV. Specify:							

VI. Model Explanation				
Model Interpret-ability		Reported model interpretability as intrinsic property		Described tradeoff between interpretability and complexity
		Explained limitations of low model interpretability		Compared the methods with different interpretabilities
Model Explain-ability		Reported the explanation method(s) used for FIA	Explanation method(s):	
		Reminded high accuracy might not be trustable explanation		Described application of explanation on the optimal model
	Reflected on understanding that explanations:		couldn't cover all mechanisms	might be incorrect
Causality		Reflected on higher accuracy not implying logical causal relationships		
		Reported understanding that explanation results do not indicate causality		
		Described alignment of explanations with environmental domain knowledge for causality		
		Identified illogical or counterintuitive parts based on environmental domain knowledge		
		Described study of illogical or counterintuitive parts based on environmental domain knowledge		

VII. Data Leakage Management				
General/ Data Source	Verified no:	response variable Y leakage	future-to-current data leakage	current-to-current data leakage
	Confirmed that no overlapped data between training and testing	Yes	No	
	Data splitting method for literature-source data:			
Data Enrichment	Verified that missing data replacement utilized only seen dataset	N/A	Yes	No
	Verified that missing data interpolation utilized neighbor data in same subset	N/A	Yes	No
	Verified that model-based imputation utilized only seen data	N/A	Yes	No
Feature Engineering	Confirmed that feature engineering utilized only seen dataset	N/A	Yes	No
	Confirmed that feature scaling utilized only seen dataset	N/A	Yes	No
	Confirmed that data splitting method for time-dependent data:			
CV Loop	Verified that missing data replacement utilized only seen data in each split	N/A	Yes	No
	Verified that missing data interpolation with neighbor data in each split	N/A	Yes	No
	Verified that feature engineering utilized only seen data in each split	N/A	Yes	No
	Verified that feature scaling utilized only seen data in each split	N/A	Yes	No
	Data splitting method for literature-source data in CV loop:			
	Data splitting method for time-dependent data in CV loop:			
Others	Self-specify:			

VIII. Additional Items		
	PURL or DOI are publicly available. Data:	Code:
	Email or URL are available to request. Data:	Code:
	Self-specified Item:	